ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

COMPUTER NETWORKS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BSc Computer Science, 3rd year, semester 1 final examination: network models and delay, the link layer and error detection, IPv4 addressing, subnetting and routing, the transport layer (TCP and UDP), and application-layer protocols and security.

Written by Academic Central for practice, following third-year Computer Science syllabuses at South African universities (computer networks, top-down approach). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SIX questions. Answer ALL the questions.
2. Show ALL your working in calculations. Give units with your answers.
3. Non-programmable calculators may be used.
4. Number your answers exactly as the questions are numbered.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which layer is responsible for routing packets from the source host to the destination host across several networks?

A Link layer
B Network layer
C Transport layer
D Application layer

1.2 How many bits are there in an IPv6 address?

A 32
B 48
C 64
D 128

1.3 Which protocol finds the MAC address that belongs to an IPv4 address on the same LAN?

A DNS
B DHCP
C ARP
D ICMP

1.4 The well-known server port number for HTTPS is …

A 21
B 80
C 443
D 25

1.5 Which device forwards frames using MAC addresses and a table that it learns by itself?

A Hub
B Switch
C Router
D Repeater

1.6 The ping command sends messages of which protocol?

A ICMP
B UDP
C TCP
D ARP

1.7 Which of these is a private IPv4 address?

A 172.32.5.1
B 192.168.10.4
C 8.8.8.8
D 100.1.1.1

1.8 In TCP, the receive window that the receiver advertises is used for …

A congestion control.
B flow control.
C error detection.
D routing.

1.9 Which protocol does a laptop use to get an IP address automatically when it joins a Wi-Fi network?

A DNS
B ARP
C DHCP
D SMTP

1.10 What is the largest number of usable host addresses in a /26 subnet?

A 62
B 64
C 30
D 126

[10]

SECTION B

QUESTION 2: NETWORK MODELS AND DELAY

2.1 Name the FIVE layers of the Internet (TCP/IP) protocol stack, from the top down. Give ONE protocol or technology that works at each layer. (5)

2.2 Explain what encapsulation is as data moves down the protocol stack at the sending host. (2)

2.3 Distinguish between circuit switching and packet switching. Give ONE advantage of each. (4)

2.4 A packet of 1 500 bytes is sent over a single link with a transmission rate of 10 Mbps. The link is 2 000 km long and the signal travels at 2×10^8 m/s.

2.4.1 Calculate the transmission delay. (2)

2.4.2 Calculate the propagation delay. (1)

2.4.3 Ignoring processing and queuing delays, calculate the total delay for the packet to reach the other end. (1)

[15]

QUESTION 3: THE LINK LAYER

3.1 A sender uses EVEN parity on the 7-bit data word 1011001.

3.1.1 Give the parity bit that is added. (1)

3.1.2 Explain why a single parity bit cannot detect an error in which TWO bits are flipped. (2)

3.2 Data 1101011 is sent using a cyclic redundancy check (CRC) with the generator G = 1011 ($x^3 + x + 1$). Use modulo-2 division to find the CRC bits, and write down the full frame that is transmitted. (5)

3.3 Explain how CSMA/CD works on a shared Ethernet, and why it is not needed on modern switched, full-duplex Ethernet. (4)

3.4 A switch has just been switched on, so its switch table is empty. Host A on port 1 sends a frame to host B. Explain what the switch does with the frame, and what it learns. (3)

[15]

QUESTION 4: THE NETWORK LAYER

4.1 An organisation is given the address block 196.21.40.0/24. It must divide the block into FOUR subnets of equal size.

4.1.1 Give the subnet mask of each subnet in prefix (/n) form and in dotted-decimal form. (2)

4.1.2 For EACH subnet, give the network address, the broadcast address and the range of usable host addresses. (8)

4.1.3 How many usable host addresses does each subnet have? (1)

4.2 A router uses longest-prefix matching with this forwarding table:

Destination prefix	Interface
10.1.0.0/16	0
10.1.4.0/22	1
10.1.5.0/24	2
Default (0.0.0.0/0)	3

Give the interface used for each destination address: (a) 10.1.5.77 (b) 10.1.6.9 (c) 10.1.200.3 (d) 10.2.0.1 (4)

4.3 FIGURE 1 shows a network of six routers. The numbers are link costs. Use Dijkstra's link-state algorithm to find the least-cost path from router u to every other router. Show a table with the distance D(x) and predecessor p(x) of each router at every step, and then write down the forwarding table of router u. (7)

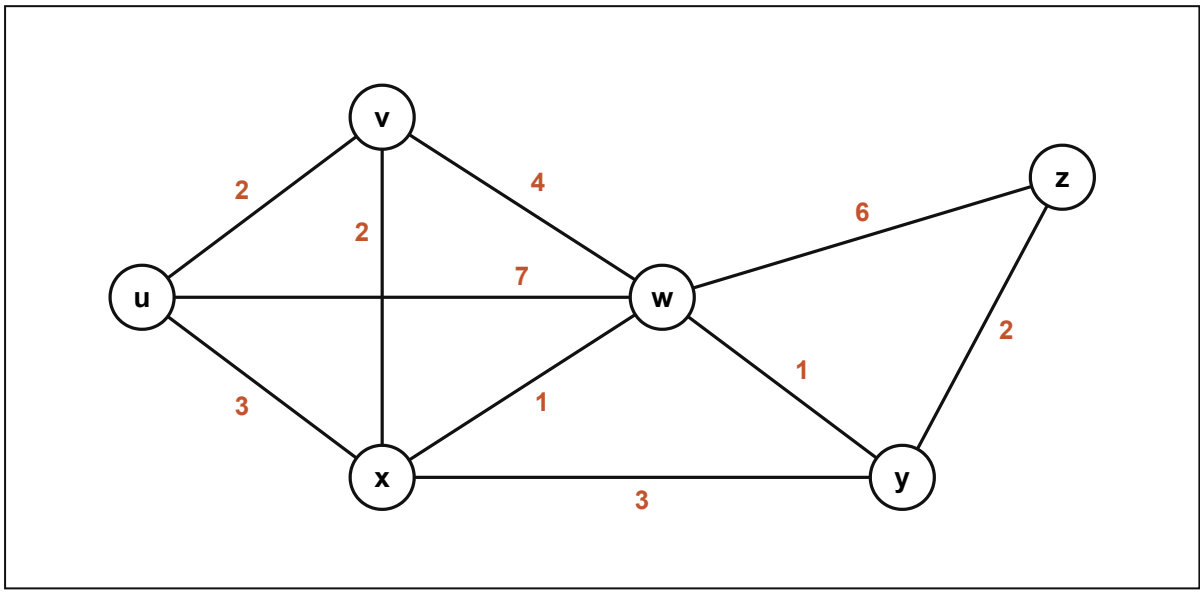


FIGURE 1: Network of routers with link costs

4.4 Explain the main difference between link-state routing and distance-vector routing. Name the problem in which distance-vector routers slowly increase their costs after a link fails. (3)

[25]

QUESTION 5: THE TRANSPORT LAYER

5.1 Give THREE differences between TCP and UDP. (3)

5.2 Name ONE application that usually runs over UDP, and explain why UDP suits it. (2)

5.3 A client opens a TCP connection to a server. The client chooses the initial sequence number 100 and the server chooses 300. Show the THREE segments of the three-way handshake, giving for each segment the flags that are set, the sequence number and (where used) the acknowledgement number. (4)

5.4 Host A sends two TCP segments to host B, one straight after the other. The first has sequence number 92 and carries 8 bytes of data; the second has sequence number 100 and carries 20 bytes.

5.4.1 The first segment arrives. Which acknowledgement number does B send? (1)

5.4.2 The first segment is lost and the second arrives. Which acknowledgement number does B send? (1)

5.4.3 Explain how TCP fast retransmit works. (2)

5.5 A TCP Reno connection starts in slow start with cwnd = 1 MSS and ssthresh = 16 MSS. No segments are lost at first. One transmission round lasts one RTT.

5.5.1 Give cwnd (in MSS) at the start of transmission rounds 1 to 7. (3)

5.5.2 During round 7, three duplicate ACKs arrive. Give the new values of ssthresh and cwnd. (Ignore the three extra MSS that are added during fast recovery.) (2)

5.5.3 If a timeout had happened in round 7 instead, what would ssthresh and cwnd be, and which phase would TCP be in? (2)

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QUESTION 6: APPLICATIONS AND SECURITY

6.1 A user types www.academiccentral.co.za into a browser. The local DNS server has nothing in its cache. Explain, step by step, how DNS finds the IP address using iterative queries. (4)

6.2 A web page consists of one base HTML file and 4 images, all on the same server. Ignore transmission times.

6.2.1 Explain the difference between non-persistent and persistent HTTP connections. (2)

6.2.2 How many RTTs are needed to fetch the whole page with (a) non-persistent HTTP with no parallel connections, and (b) persistent HTTP without pipelining? (2)

6.3 Name the protocol used (a) to send an email from a mail app to the sender's mail server, and (b) to read email that stays on the server from both a phone and a laptop. (2)

6.4 Explain the difference between symmetric-key and public-key cryptography, and explain how TLS uses both. (3)

6.5 Explain what a stateful firewall can do that a simple packet filter cannot. (2)

[15]

TOTAL: 100

END OF QUESTION PAPER